Osaid Khan

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Education

Pace University New York City, NY
Master of Science (MS) in Computer Science (GPA - 4.0/4.0) *Jan 2024 - Dec 2025*

Dr. A.P.J. Abdul Kalam Technical University Noida, India
Bachelor of Technology (B.Tech) in Computer Science *Jul 2016 - Sep 2020*

Technical Skills

Systems & Backend: *Rust*, Python, Java, TypeScript, *PostgreSQL*, *Redis*, *Kafka*, *Distributed Systems*, *CI/CD*
Cloud & Infrastructure: AWS (*Step Functions*, *Lambda*, *S3*, *RDS*, *EC2*), Google Cloud, Git, Node.js
Frameworks & AI: LangGraph, LangChain, FastAPI, FastMCP, LangSmith, PyTorch, React, Spring Boot

Professional Experience

Software Engineer Phoenix, AZ
Sperse *Feb 2026 - present*

- Architected a **concurrent multi-agent orchestration system** using **LangChain** and **LangGraph**, implementing **DAG workflow primitives** for scalable tool execution and agent coordination
- Developed an **agentic interaction layer** in **Rust** and **TypeScript** that translates natural-language intents into **idempotent, validated tool calls** across **1000+ platform endpoints**
- Optimized concurrency and prompt routing to maximize **KV-cache hit rate**, implementing **backpressure** and **scheduling** to reduce end-to-end **latency** by **60%** and lower operational costs

AI Researcher New York City, NY
Pace Artificial Intelligence Lab *Sep 2025 - Dec 2025*

- Engineered high-performance training pipelines for **StyleGAN** and **Stable Diffusion** using **CUDA** and **PyTorch**, leveraging an **HPC cluster** for model fine-tuning and **systems-level optimization**
- Optimized model **inference** and real-time delivery using **vLLM**, ensuring secure, **scalable model serving** with minimal **latency** for production-grade AI applications

Software Engineer Noida, India
Qualitest *Feb 2021 - Dec 2023*

- Designed **durable backend workflows** using **AWS Step Functions** to orchestrate **long-running job execution**, ensuring **reliable state management** and **idempotency**
- Built a **cloud-agnostic storage engine** using **Python**, **PostgreSQL**, and **Redis** to handle **distributed file operations** across **S3**, **GCS**, and **MinIO** via **streaming** and signed URL patterns

Projects

Dark Factory - Orchestrator for AI coding CLIs | *Crossterm, Ratatui, Y-crdt* github.com/ozzyozbourne/dark-factory

- Built a **distributed execution orchestrator** using **Rust** that enables full ownership of **CLI IO via PTY**, managing **real-time event-driven loops** with **Crossterm** and **Ratatui**
- Implemented a **streaming** signaling layer using **Y-crdt** to coordinate **agentic loops** across machines, eliminating **polling overhead** and ensuring **reliable state synchronization** in **distributed systems**